Practice Set B

Class 12 Mathematics, Chapter 3: Matrices

Maximum marks: **49** **20** questions

Name _____ Date _____ Score _____ / 49

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1 — 8 marks

1 — MCQ, PYQ-style — 1 mark

If A is a matrix of order $m \times n$ and B is a matrix such that AB' and $B'A$ are both defined, then the order of B is:

(A) $m \times m$ **(B)** $n \times n$ **(C)** $n \times m$ **(D)** $m \times n$

2 — MCQ, PYQ-style — 1 mark

If $\begin{bmatrix} x+y & 2 \\ 5 & xy \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix}$, then the value of $\left(\frac{24}{x} + \frac{24}{y}\right)$ is:

(A) 3 **(B)** 4 **(C)** 6 **(D)** 18

3 — MCQ, NCERT Ex 3.3 Q11 — 1 mark

If A and B are symmetric matrices of the same order, then $AB - BA$ is a:

(A) skew-symmetric matrix **(B)** symmetric matrix

(C) zero matrix **(D)** identity matrix

4 — MCQ, Board-pattern — 1 mark

Let $A = [a_{ij}]$ be a 3×2 matrix whose elements are given by $a_{ij} = (i - 3j)^2$. The element in the 2nd row and 3rd column of A' is:

(A) 0 **(B)** 9 **(C)** 16 **(D)** 49

5 — MCQ, NCERT Exemplar — 1 mark

If A is a square matrix such that $A^2 = I$, then $(A - I)^3 + (A + I)^3 - 7A$ is equal to:

(A) A **(B)** $I - A$ **(C)** $I + A$ **(D)** $3A$

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** For $A = \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix}$, $A^2 = O$. **Reason (R):** If the product of two matrices is a zero matrix, then at least one of the two matrices must be a zero matrix. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, Board-pattern Find the values of x and y if $2\begin{bmatrix} x & 3 \\ y & x \end{bmatrix} - \begin{bmatrix} 1 & y \\ x & 4 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 5 & 2 \end{bmatrix}$.	2 marks

Level 2

18 marks

8	Short answer, Board-pattern Construct a 3×3 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = \begin{cases} i - j, & \text{if } i \geq j \\ i + j, & \text{if } i < j \end{cases}$	2 marks
9	Short answer, PYQ-style For the matrix $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$, find the numbers a and b such that $A^2 + aA + bI = O$. Hence find A^{-1}.	3 marks
10	Short answer, Board-pattern Express the matrix $A = \begin{bmatrix} 2 & 5 & -3 \\ 1 & 4 & 6 \\ 7 & -2 & 0 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.	3 marks
11	Short answer, Board-pattern Find the matrices X and Y if $X + 2Y = \begin{bmatrix} 4 & 7 \\ -4 & 5 \end{bmatrix}$ and $2X - Y = \begin{bmatrix} 3 & -6 \\ 2 & 5 \end{bmatrix}$.	2 marks
12	Short answer, Board-pattern Find x, if $\begin{bmatrix} x & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ x \\ 2 \end{bmatrix} = O$.	3 marks
13	Short answer, Board-pattern If $A = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$, find $\alpha \left(0 < \alpha < \frac{\pi}{2}\right)$ such that $A^2 + I = \sqrt{3}\,A$.	2 marks
14	Short answer, Board-pattern If $A' = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$, verify that $(AB)' = B'A'$.	3 marks

Q No.	Question	Marks

Level 3 — 23 marks

15 — Long answer, PYQ-style — 3 marks

If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, show that $A^2 - 4A - 5I = O$. Hence find A^{-1}.

16 — Long answer, Board-pattern — 3 marks

A stationery shop sells gel pens, ball pens and fountain pens. On Day 1 it sold 20 gel pens, 30 ball pens and 10 fountain pens. On Day 2 it sold 15 gel pens, 40 ball pens and 12 fountain pens. A ball pen costs ₹10 and a fountain pen costs ₹50. The total sales of the two days were ₹2,850. Using matrix multiplication, find the price of a gel pen and the sales on each day.

17 — Long answer, PYQ-style — 3 marks

If $A = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$, find $A^2 - 5A - 14I$. Hence obtain A^3.

18 — Long answer, PYQ-style — 5 marks

If $A = \begin{bmatrix} \cos^2\alpha & \cos\alpha\sin\alpha \\ \cos\alpha\sin\alpha & \sin^2\alpha \end{bmatrix}$ and $B = \begin{bmatrix} \cos^2\beta & \cos\beta\sin\beta \\ \cos\beta\sin\beta & \sin^2\beta \end{bmatrix}$, show that AB is the zero matrix when α and β differ by an odd multiple of $\dfrac{\pi}{2}$.

19 — Long answer, PYQ-style — 5 marks

If $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix}$, show that $A^3 = pI + qA + rA^2$.

20 — Case study, Board-pattern — 4 marks

Case Study: A dairy cooperative sells milk, curd and paneer in two towns P and Q. In one week, town P buys 500 litres of milk, 150 kg of curd and 40 kg of paneer; town Q buys 400 litres of milk, 200 kg of curd and 50 kg of paneer. The selling prices are ₹60 per litre of milk, ₹80 per kg of curd and ₹400 per kg of paneer. The cost prices are ₹45, ₹60 and ₹320 respectively.

(i) Write the weekly sales as a 2×3 matrix and the selling prices as a column matrix. — 1 mark

(ii) Find the revenue from town P. — 1 mark

(iii)(a) Using matrix multiplication, find the profit from each town and the total profit. — 2 marks

OR

(iii)(b) The cooperative changes the price of paneer to ₹x per kg, keeping the other prices the same. The total revenue from both towns becomes ₹1,22,500. Using matrices, find x. — 2 marks

Answer key

Chapter 3: Matrices, Practice Set B. Full step-wise solutions are in the solutions sheet.

Level 1

1	(D) $m \times n$
2	(D) 18
3	(A) skew-symmetric matrix
4	(B) 9
5	(A) A
6	(C) Assertion (A) is true, but Reason (R) is false.
7	$x = 3, y = 4$

Level 2

8	$A = \begin{bmatrix} 0 & 3 & 4 \\ 1 & 0 & 5 \\ 2 & 1 & 0 \end{bmatrix}$
9	$a = -4, b = 1$; $A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$
10	$A = \begin{bmatrix} 2 & 3 & 2 \\ 3 & 4 & 2 \\ 2 & 2 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 2 & -5 \\ -2 & 0 & 4 \\ 5 & -4 & 0 \end{bmatrix}$
11	$X = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}, Y = \begin{bmatrix} 1 & 4 \\ -2 & 1 \end{bmatrix}$
12	$x = \dfrac{2}{3}$ or $x = -1$
13	$\alpha = \dfrac{\pi}{6}$
14	$(AB)' = B'A' = \begin{bmatrix} 2 & -3 & 5 \\ -1 & 5 & -6 \end{bmatrix}$, hence verified.

Level 3

15	$A^2 - 4A - 5I = O$; $A^{-1} = \dfrac{1}{5}\begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$
16	Gel pen ₹30; Day 1 sales ₹1,400; Day 2 sales ₹1,450
17	$A^2 - 5A - 14I = O$; $A^3 = \begin{bmatrix} 187 & -195 \\ -156 & 148 \end{bmatrix}$
18	$AB = \cos(\alpha - \beta)\begin{bmatrix} \cos\alpha\cos\beta & \cos\alpha\sin\beta \\ \sin\alpha\cos\beta & \sin\alpha\sin\beta \end{bmatrix} = O$ when $\alpha - \beta = (2n+1)\dfrac{\pi}{2}$
19	Both sides equal $\begin{bmatrix} p & q & r \\ pr & p+qr & q+r^2 \\ pq+pr^2 & pr+q^2+qr^2 & p+2qr+r^3 \end{bmatrix}$, so $A^3 = pI + qA + rA^2$.
20	Revenue from P ₹58,000. Profit: P ₹13,700, Q ₹14,000, total ₹27,700. OR: $x = 450$.